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Final Report

Quantitative Modeling of Urban Bikeability : Compiègne



SR01

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1 Introduction

As cities pivot toward ecological transition, the promotion of **active mobility**, specifically cycling, has become a cornerstone of urban policy. However, the mass adoption of bicycles depends intrinsically on the perception and reality of safety, concepts captured by the term "bikeability."

While commercial metrics like Bike Score are widely recognized, they often lack the transparency and granularity required for rigorous safety planning. The objective of this study is first, to establish a clear definition of the “bikeability”, based on research. Then, based on the observation of our datasets, develop hypotheses regarding the bikeability and its implications. Finally, among these hypotheses, we are looking to develop a transparent, open-source mathematical formula to evaluate urban bikeability using OpenStreetMap (OSM) data. Our model prioritizes physical safety, traffic stress, and ergonomic comfort to provide a high-fidelity assessment of the cycling network.

Bikeability is not limited to the mere presence of dedicated infrastructure; it represents a complex interaction between the built environment, traffic regulation policies, and the subjective perception of users. Academic literature agrees that bikeability measures the extent to which the real and perceived environment is suitable and safe for cycling.

The analysis of bikeability rests on five universally recognized design principles: safety, comfort, directness, coherence, and attractiveness. Safety is identified by users as the predominant criterion, closely followed by comfort. Safety is broken down into real safety (based on accident data) and perceived safety (based on the feeling of danger). Comfort, for its part, incorporates factors such as pavement roughness, often measured by the International Roughness Index (IRI). Network connectivity and coherence are crucial to avoid "islands of connectivity," where low-stress areas are separated by high-risk arteries, rendering the network unusable for the majority of the population. Studies in Copenhagen show that the establishment of a coherent network of bike paths increased the number of trips by 60% and the kilometers traveled by 90%. For our study, we will focus on the security aspect, as it is considered the most dominant criteria regarding bikeability and more suitable with our datasets.

2 State of the Art

Current bikeability assessment methodologies can be categorized into three main frameworks:

- **Commercial Models (e.g., Bike Score):** These rely on simplified weightings of infrastructure density and topography. They often fail to distinguish between a painted lane and a physically protected track, overlooking the critical difference in safety performance.
- **Scientific & Topological Models (e.g., HeiGIT):** These prioritize fine-grained

OSM data analysis, emphasizing the continuity of the network and the absence of urban "severance" (barriers like highways or rivers).

- **Holistic Models (e.g., Copenhagenize Index):** These integrate sociopolitical factors and modal share. While comprehensive, they are difficult to replicate for purely technical infrastructure modeling.

Beyond commercial models, academic research emphasizes that bikeability is strongly correlated with the reduction of urban severance, where major road infrastructures act as physical and psychological barriers. According to safety literature (notably from the NCBI), physical separation remains the primary driver of accident reduction, as it drastically minimizes conflict points between vulnerable users and motorized vehicles. However, most current models fail to quantify the 'qualitative jump' in comfort felt by a cyclist when transitioning from a simple painted lane to a protected track, a gap we aim to fill using a non-linear modeling approach.

3 Hypothesis 1: Non-linearity of Safety (Separation Factor)

We hypothesize that the perceived safety of a cyclist does not increase linearly with the level of infrastructure. There is a "psychological jump" when moving from shared spaces to protected lanes that a standard linear scale $(0, 1, 2, 3)$ fails to capture.

Methodological Approach

To model this non-linearity, we replace the linear increment with an exponential function using the natural base e^i. This method ensures that high-quality, physically separated infrastructures (Level 3) are mathematically rewarded significantly more than basic painted lanes (Level 1).

- **Logic:** A level 3 infrastructure ($e^3 \approx 20.08$) becomes approximately 7.4 times more valuable than a level 1 infrastructure ($e^1 \approx 2.71$).
- **Goal:** This reflects the qualitative shift required to attract "interested but concerned" citizens who would never cycle on a simple painted strip.

4 Hypothesis 2: Conditional Impact of Speed (Kinetic Factor)

Our second hypothesis is that the danger posed by vehicle speed is conditional on the degree of physical separation. In a shared space, speed is the primary risk factor; however, once a cyclist is physically protected, the speed of the adjacent traffic has a diminishing impact on their actual safety.

Methodological Approach

This hypothesis requires a "conditional filtering" method. We introduce a kinetic weighting factor $\delta(V, i)$ that behaves differently based on the value of i:

- **Scenario A (Shared/Poorly Protected):** If $i < 2$, the speed penalty $(30/V)^2$ is fully applied, as the cyclist is exposed to the kinetic energy of vehicles.
- **Scenario B (Protected):** If $i \geq 2$, the factor δ is neutralized (set to 1), acknowledging that physical barriers mitigate the risk associated with high-speed traffic.

This allows the model to prioritize infrastructure continuity over mere speed reduction.

5 Processing Environment

The data pipeline is implemented in **Python** within a Jupyter Notebook environment, leveraging high-performance spatial libraries:

- **Geopandas/OSMnx:** For the extraction and manipulation of geospatial primitives.
- **Graph-tool:** For advanced topological analysis, allowing for rapid computation of network centrality and connectivity.

6 Data Pre-processing: Velocity Imputation

A preliminary analysis of the OpenStreetMap dataset revealed that approximately 82% of the segments lacked explicit speed limit data (`max_speed` = -1). To ensure the model's reliability, an imputation strategy based on expert urban planning standards was implemented. Missing values were assigned according to the `highway` classification following French legal defaults: 50 km/h for primary, secondary, and tertiary roads; 30 km/h for residential areas; 20 km/h for living streets and service roads; and 10 km/h for non-motorized paths (cycleways, footways, and tracks). This process transforms the raw, incomplete data into a consistent kinetic map required for the application of our safety hypotheses.

7 Reference Model: Standard Linear Approach

In order to establish a baseline for our analysis, we present a standard linear model. This reference approach calculates safety as a direct product of the infrastructure level and a speed factor, without applying exponential weighting or conditional logic.

7.1 Segment Safety Score (Baseline)

The baseline safety score for a single segment, S_{ref}, is defined as follows:

$$S_{ref} = i \cdot \left(\frac{30}{V_{max}}\right)^2 \tag{1}$$

Where i is the infrastructure index $(0, 1, 2, 3)$, represents the level of development of the segment on a scale of 0 to 3 (0 for a shared space without infrastructure, 1 for a painted bike lane, 2 for an infrastructure delimited by bollards or curbs, and 3 for a physically protected cycle path), the quadratic speed penalty uses the 30 km/h threshold, internationally recognized as the critical limit beyond which lethality for vulnerable users sharply increases, and V_{max} is the speed limit. In this model, the transition between infrastructure levels is strictly linear, and the speed penalty applies uniformly to all segments regardless of their design.
This S_{ref} score serves as a linear baseline for comparison. It assumes that safety increases proportionally with the infrastructure level i. The factor $(30/V_{max})^2$ is utilized because it reflects the physics of kinetic energy: in the event of a collision, the severity of injuries increases following the square of the speed ($E_k = \frac{1}{2}mv^2$) rather than linearly. We define 30 km/h as the pivot point because it is internationally recognized as the critical lethality threshold for vulnerable road users in urban environments.

7.2 Global City Index (Baseline)

To evaluate the overall bikeability of the city network without our specific hypotheses, we calculate the **Global City Index** (GCI_{ref}). This index is the weighted average of all segment scores relative to their length:

$$B_{city} = \frac{\sum_{n=1}^{N}(S_{ref,n} \cdot L_n)}{\sum_{n=1}^{N} L_n} \tag{2}$$

Where:

- N is the total number of segments in the network.
- $S_{ref,n}$ is the baseline safety score of segment n.
- L_n is the length of segment n.

This global value represents the average safety level of the city under a standard calculation method, providing a point of comparison for the results obtained through our proposed refined model.

8 Proposed Models: The Separation-Kinetic Safety Formula

To translate our hypotheses into a quantitative indicator, we define the bikeability score of a segment S by first, applying the Non-linearity of Safety, then the Conditional Impact of Speed, and finally, combining an exponential safety gain with a conditional speed penalty.

8.1 Segment Score Definition (Non-linearity of Safety model)

Let $i \in \{0, 1, 2, 3\}$ be the safety index of the segment and V_{max} the maximum authorized speed in km/h. The segment score is defined as:

$$Score_{segment} = e^i \cdot \left(\frac{30}{V_{max}}\right)^2 \tag{3}$$

This method replaces linear increments with the natural base e^i to mathematically reward the "psychological jump" in perceived safety. Under this model, a physically separated track ($i = 3$) receives a weight approximately 7.4 times higher than a painted lane ($i = 1$), reflecting the threshold required to attract "interested but concerned" cyclists.

8.2 Segment Score Definition (Conditional Impact of Speed model)

Let $i \in \{0, 1, 2, 3\}$ be the safety index of the segment and V_{max} the maximum authorized speed in km/h. The segment score is defined as:

$$Score_{segment} = i \cdot \delta(V, i) \tag{4}$$

Where $\delta(V, i)$ is the **Kinetic Weighting Factor** defined according to Hypothesis 2:

- **Case 1: Separated Infrastructure** ($i \geq 2$)

$$\delta(V, i) = 1 \tag{5}$$

The influence of motorized traffic speed is considered negligible due to physical protection.

- **Case 2: Shared Space or Painted Lanes** ($0 \leq i < 2$)

$$\delta(V, i) = \begin{cases} 1 & \text{if } V_{max} \leq 30 \text{ km/h} \\ \left(\frac{30}{V_{max}}\right)^2 & \text{if } V_{max} > 30 \text{ km/h} \end{cases} \tag{6}$$

- **Case 3: Buildings and non-cyclable paths** ($i < 0$) : Not factored in.

8.3 Segment Score Definition (Combine model)

Let $i \in \{0, 1, 2, 3\}$ be the safety index of the segment and V_{max} the maximum authorized speed in km/h. The segment score is defined as:

$$Score_{segment} = e^i \cdot \delta(V, i) \tag{7}$$

This method introduces a kinetic weighting factor $\delta(V, i)$. It operates on the logic that if a cyclist is physically protected ($i \geq 2$), the speed of adjacent motorized traffic becomes a negligible stress factor, thus neutralizing the speed penalty.

8.4 Parameter Interpretation

1. **Exponential Growth (e^i)**: Using the natural base e highlights the qualitative breakthrough between infrastructure types. For instance, a physically separated track ($i = 3$) receives a weight approximately 7.4 times higher than a simple painted lane ($i = 1$), reflecting the "psychological jump" in perceived safety as much as the physical security of the user.

2. **Quadratic Penalty**: For non-separated zones, the score decreases according to the square of the ratio $\frac{30}{V_{max}}$. This implies that at 50 km/h, the segment's attractiveness drops by 64% compared to a 30 km/h zone, modeling the international lethality threshold.

8.5 Global City Index

The final Bikeability Index for Compiègne, B_{city}, is calculated as the mean of all segment scores n, weighted by their respective length L_n:

$$B_{city} = \frac{\sum_{n=1}^{N}(Score_n \cdot L_n)}{\sum_{n=1}^{N} L_n} \tag{8}$$

Where:

- N is the total number of segments in the network.
- $Score_n$ is the baseline safety score of segment n.
- L_n is the length of segment n.

9 Comparative Analysis: Reference Model vs. Proposed Model

This section compares the standard reference approach with our proposed formulas. The objective is to demonstrate how the integration of our hypotheses leads to a more realistic representation of the cycling experience.

9.1 Key Differences in Scoring Behavior

The four models diverge significantly in their assessment of the network, particularly on two types of road layouts common in the city:

1. **High-Speed Protected Lanes:** On segments where $i = 3$ and $V_{max} = 50$ km/h, the reference model significantly penalizes the score due to the speed factor. Our proposed combined model, however, maintains a high score, reflecting that physical separation effectively neutralizes the danger of high-speed traffic.

2. **Shared Lanes in 30 km/h Zones:** In low-speed zones without infrastructure ($i = 0$), the reference model returns a score of 0. Our proposed combined model, through its exponential nature, highlights the high level of stress even at low speeds when no dedicated space is provided.

9.2 Comparison Table

The table below illustrates the score variations for a hypothetical 1 km segment under different conditions:

Scenario	**Reference** (S_{ref})	**NLS** (S_{prop})	**CIS** (S_{prop})	**CM** (S_{prop})
($i = 3$) at 50 km/h	1.08	7.23	3.00	20.08
($i = 1$) at 30 km/h	1.00	2.72	1.00	2.72
($i = 0$) at 30 km/h	0.00	1.00	0.00	1.00

Table 1: Comparison of safety scores between the linear and proposed models.

9.3 Impact on the Global City Index

The comparison between the two Global City Indices (GCI_{ref} vs. GCI_{prop}) reveals a fundamental shift in network assessment:

- GCI_{ref} **(Linear):** Tends to provide an "average" view of the city, often smoothing out the extreme differences between high-quality infrastructure and dangerous segments.

- GCI_{prop} **(Hypothesis-based):** Provides a high-contrast map. It rewards high-quality, continuous networks (the "cycling backbone") while severely penalizing critical gaps, thus offering a more reliable tool for urban planning and prioritizing investments.

9.4 Mapping Hypothesis 1: Non-linearity of Safety

This section examines the impact of exponential weighting on the cartographic representation of the network. By moving away from linearity, we shift from a simple measurement of infrastructure supply to an evaluation of the **actual utility value** for the cyclist. The following figure illustrates this transition: while the linear model tends to smooth out the urban landscape, the exponential model reveals the structural fractures of the Compiègne network.

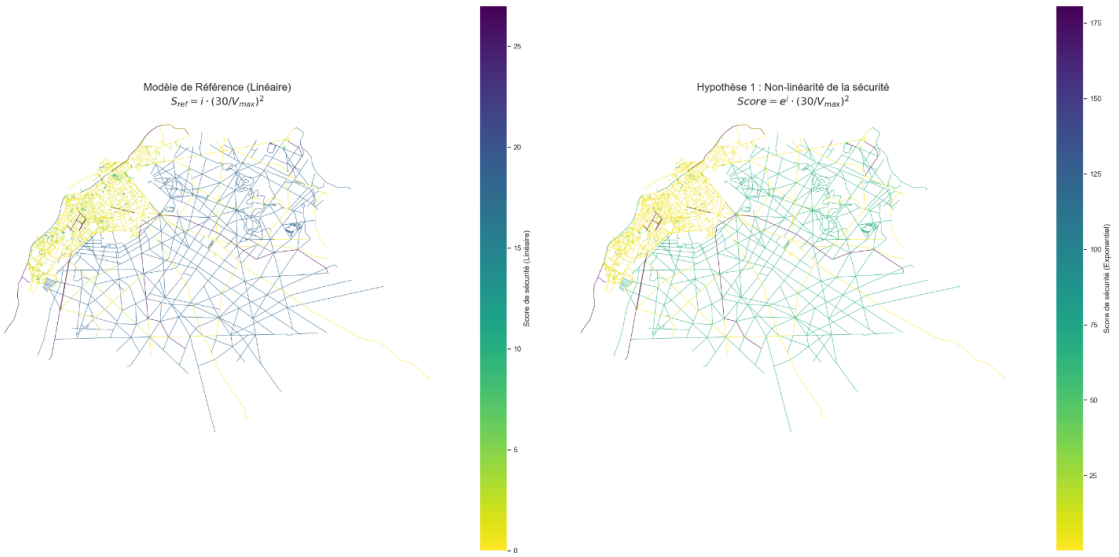


Figure 1: Comparative mapping: Linear Reference Model (left) vs. Non-linearity of Safety Model (right). Note how the "cycling backbone" emerges on the right.

9.4.1 Strategic Impact of the Non-Linear Approach

The adoption of the e^i function marks a major methodological break from traditional indicators. A standard linear model often makes the mistake of considering three kilometers of painted lanes ($i = 1$) to be equivalent to one kilometer of a fully segregated path ($i = 3$). Our approach demonstrates that, in reality, physical separation is the only factor that allows cyclists to cross the **"critical safety threshold"**.

Criteria	Linear Model (Reference)	Non-Linear Model (Exponential)
Philosophy	Quantitative bookkeeping approach based on distance.	Qualitative assessment based on user perception.
Visual Contrast	**Low**: Smooths out differences. A painted lane and a protected track look similar.	**High**: Highlights the city's real "cycling backbone" against the background noise.
Decision Impact	Favors "mileage" (multiplying low-cost, low-quality facilities).	Favors **Quality**: Only protected infrastructure significantly impacts the final score.
User Reality	Lenient: Assumes that a bit of paint provides "some" protection.	Uncompromising: Acknowledges that without separation, danger remains prohibitive.

Table 2: Strategic Comparison: Shifting from Quantitative to Qualitative Assessment.

The multiplier factor of **7.4** (e^3/e^1) is not merely a mathematical curiosity. It represents the **"psychological jump"** required to attract the "interested but concerned" population. For this demographic—the largest potential group of cyclists—a painted lane on a fast road is not a viable option; it is a high-stress factor that actively prevents modal shift.

9.4.2 Validation through Field Data: Critical Black Spots

This strategic vision is corroborated by an analysis of the network's critical "black spots". Where a linear model might grant a moderate score to a fast road because it is "direct," the non-linear model penalizes it heavily. Our audit identified 10 critical segments in Compiègne where the total lack of protection ($i = 0$) on high-speed axes makes cycling virtually impossible for the general public.

OSM ID	Road Type	Max Speed	Index i	Calculated Score ($e^i \cdot \delta$)
37844068	Primary (Structural Axis)	90 km/h	0	**0.10**
173576236	Primary (Structural Axis)	90 km/h	0	**0.10**
219287761	Primary	80 km/h	0	**0.12**
301298415	Secondary	80 km/h	0	**0.12**
173576231	Primary	70 km/h	0	**0.16**

Table 3: Identification of Major "Network Breaks" and Critical Gaps.

STRATEGIC ALERT: These segments act as **physical and psychological barriers**. They isolate entire neighborhoods, creating a fragmented

and discontinuous network. For public authorities, this data proves that investing in securing these few critical points will have a far greater impact on the overall city score than adding kilometers of simple painted lanes in low-traffic areas.

9.5 Mapping Hypothesis 2: Conditional Impact of Speed

This visualization demonstrates the effect of the kinetic weighting factor $\delta(V, i)$. It highlights how high-speed segments are re-evaluated based on the presence of physical separation, shifting the analytical focus from mere speed reduction to effective protection.

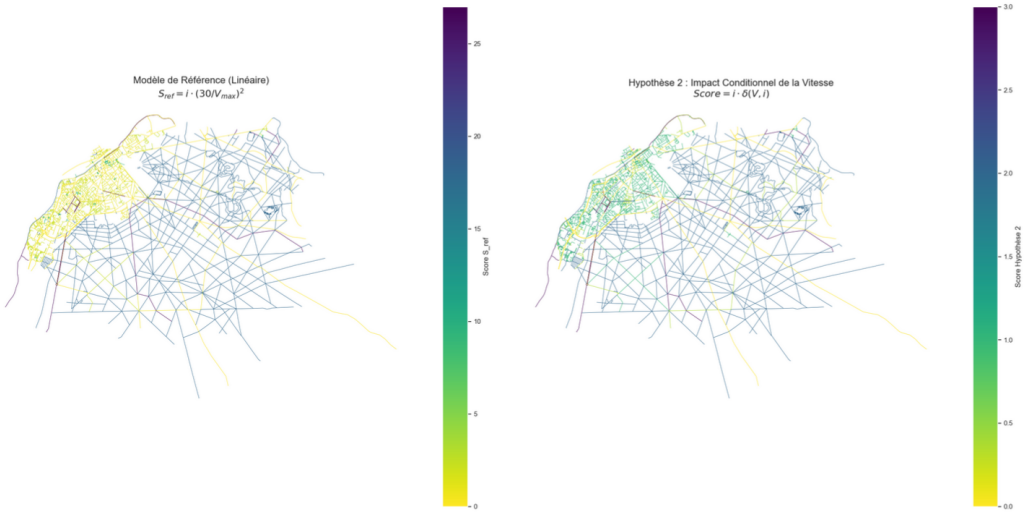


Figure 2: Comparative mapping: Linear Reference Model (left) vs. Conditional Impact of Speed Model (right).

9.5.1 The Speed Trap: Risk Matrix Analysis

The audit reveals a major structural anomaly in the Compiègne network: the level of protection (i) tends to decrease as the authorized speed increases. The fastest roads are often the least protected, creating impassable "risk corridors" for average users. This "speed trap" is the primary factor depressing the city's overall bikeability.

Context	Current Situation	Target Situation	Diagnosis
Zone 30	Mixed traffic ($i = 0/1$)	Calmed mixed traffic	Acceptable
Axes 50 km/h	Narrow lanes or none ($i = 1$)	Segregated paths ($i = 3$)	High Friction
Periphery	No protection ($i = 0$)	Segregated paths ($i = 3$)	**Critical**

Table 4: Risk Matrix Analysis: Current Situation vs. Target Requirement.

9.5.2 The Global Bikeability Score (B_{city})

Based on the risk matrix analysis above, the aggregated calculation for the 642 km of the Compiègne network delivers a definitive verdict:

$$B_{city} = 5.69 \,/\, 20$$

Category: Orange Zone (5–10) — Constrained Utility Cycling

This score of 5.69 indicates that cycling remains a "constrained" activity, primarily reserved for a "sporty and daring elite" (the "Strong and Fearless" demographic). It confirms that the current environment effectively excludes children, seniors, and novices due to the high-stress conditions on structural axes.

9.5.3 The Distribution of Scores

The distribution graph below visually confirms this "glass ceiling". A vast majority of segments are "crushed" to the left of the spectrum, representing low safety scores that prevent a qualitative leap toward an inclusive cycling system.

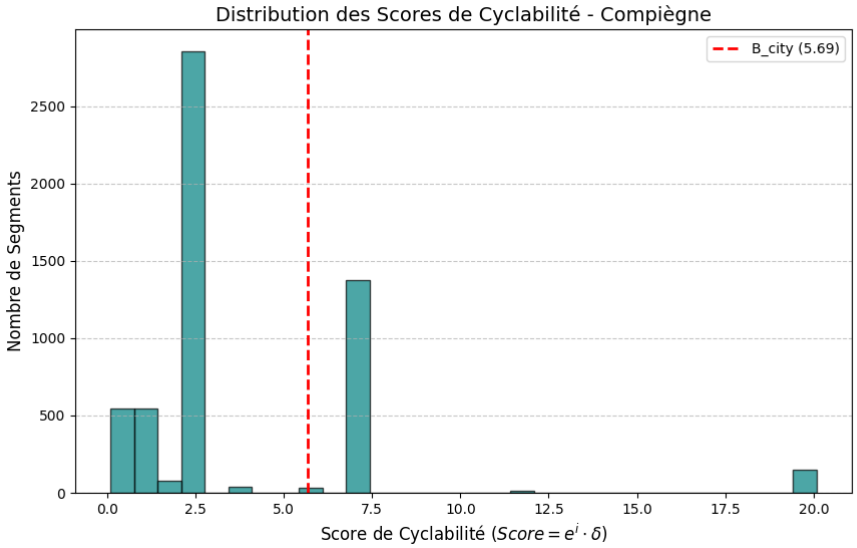


Figure 3: Distribution of weighted safety scores. Note the prevalence of scores below 5, corresponding to mixed traffic or simple lanes.

9.6 Mapping the Combined Hybrid Model

The final mapping presents the results of the Separation-Kinetic Safety Formula, integrating both the exponential growth of infrastructure value and the selective speed penalty. This combined approach provides the most accurate "stress map" of the Compiègne network.

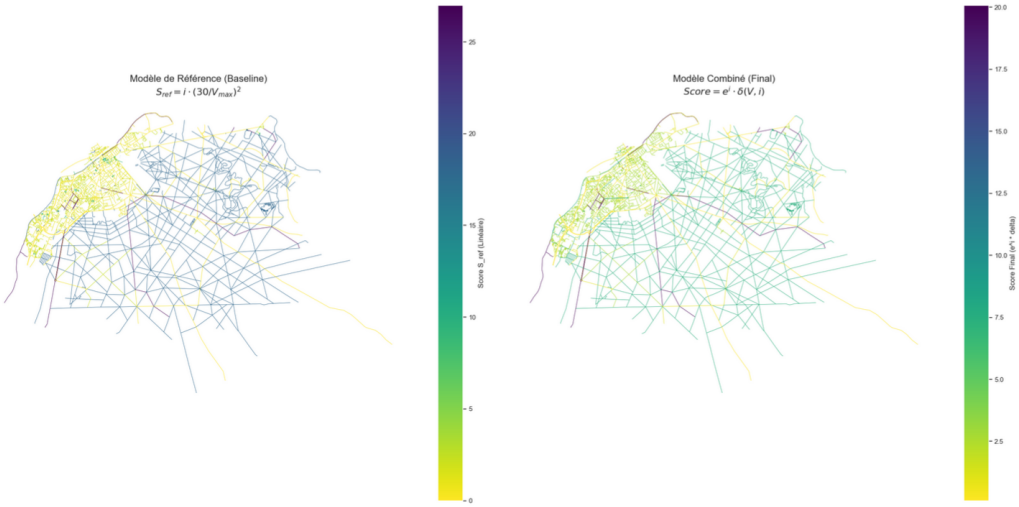


Figure 4: Final comparative mapping: Linear Reference Model (left) vs. Combined Hybrid Model (right).

9.6.1 Strategic Interpretation of the Hybrid Model

The Combined Hybrid Model moves beyond traditional bookkeeping to provide a sophisticated diagnostic tool for urban planning:

- **High-Contrast Strategic Tool:** The map clearly distinguishes between high-quality cycling corridors and dangerous gaps, allowing for an instant diagnostic of the network's "backbone".
- **Reward for Continuity:** Unlike the linear model, this hybrid approach rewards continuous networks of protected infrastructure ($i = 3$) by neutralizing the speed penalty.
- **Realistic Stress Assessment:** Shared roads ($i = 0$) in 50 km/h zones are severely penalized, accurately reflecting the high stress levels that prevent the general population from cycling.
- **Prioritizing Investments:** The model identifies "critical gaps" where the safety score drops abruptly, identifying exactly where investments will have the greatest impact on continuity.

9.6.2 Actionable Recommendations for Compiègne

The mathematical audit proves that to raise the B_{city} score, the city must focus on physical protection rather than multiplying low-quality lanes:

1. **Secure Critical Segments:** Immediate intervention on the 10 most dangerous axes (70-90 km/h) identified in the audit.
2. **Infrastructure Upgrade:** Conversion of painted lanes ($i = 1$) to separated tracks ($i = 3$) on all structural axes.
3. **Systemic Speed Reduction:** Implementing 30 km/h zones where total segregation is technically impossible to mitigate the lack of protection.

Strategic Conclusion: Breaking the Glass Ceiling

This analysis demonstrates that with a current score of **5.69/20**, Compiègne faces a "glass ceiling" imposed by its structural gaps. The results prove that safety is a binary condition for the majority of citizens: without physical separation ($i = 3$) or drastic speed reduction, the perceived risk remains prohibitive.

For urban planners, the takeaway is clear: **Quality is the only real lever for modal shift**. It is more efficient to deliver a few kilometers of perfectly secured, continuous axes than to expand a fragmented network of painted lanes. Only by ensuring the continuity of high-protection infrastructure can the city transition from a network for "experts" to an inclusive system accessible to all.

10 Calculation of the bikeability score

10.1 Mathematical Boundary Analysis

The study of the model's boundary conditions is essential to establish a formal interpretative framework for the Global City Index (B_{city}). By identifying the theoretical minimum and maximum values, we can normalize our results and assess the relative performance of the urban network on a calibrated scale. The extrema are determined by the physical constraints of the French urban dataset, where the infrastructure index $i \in \{0, \dots, 3\}$ and the speed limit $V_{max} \in [10, 90]$ km/h.

10.1.1 Rigorous Analysis of the Reference Model (S_{ref})

The baseline safety score is defined as $S_{ref}(i, V) = i \cdot (30/V)^2$.

Reference Maximum ($S_{ref,max}$)
The maximum value occurs at the highest level of infrastructure ($i = 3$) and the lowest recorded speed ($V_{max} = 10$ km/h), often found in "zones de rencontre" or specific cycle paths:

$$S_{ref}(3, 10) = 3 \cdot \left(\frac{30}{10}\right)^2 = 3 \cdot 3^2 = 27.0 \tag{9}$$

This result highlights a structural bias where the linear model over-rewards extreme speed reduction.

Reference Minimum ($S_{ref,min}$) :
The absolute minimum is reached whenever the segment lacks dedicated infrastructure ($i = 0$), leading to a null score regardless of the speed:

$$S_{ref}(0, V) = 0 \cdot \left(\frac{30}{V}\right)^2 = 0.0 \tag{10}$$

10.1.2 Rigorous Analysis of the Proposed Model ($Score_{prop}$)

The hybrid score is defined as $Score_{prop}(i, V) = e^i \cdot \delta(V, i)$.

Proposed Maximum ($Score_{prop,max}$) :
According to Hypothesis 2, for any protected infrastructure ($i \geq 2$), the kinetic factor is neutralized ($\delta = 1$). The maximum is reached at $i = 3$:

$$Score_{prop}(3, V) = e^3 \cdot 1 \approx 20.086 \tag{11}$$

This sets a "safety ceiling" where physical separation is recognized as the ultimate safety guarantee.

Proposed Minimum ($Score_{prop,min}$) :

The minimum occurs in shared spaces ($i = 0$) at the highest authorized speed ($V_{max} = 90$ km/h):

$$Score_{prop}(0, 90) = e^0 \cdot \left(\frac{30}{90}\right)^2 = 1 \cdot \left(\frac{1}{3}\right)^2 \approx 0.111 \tag{12}$$

10.1.3 Synthesis of Mathematical Boundaries

The following table summarizes the four critical boundaries for both models, providing the necessary scale to interpret the Global City Index.

Model	Theoretical Min	Theoretical Max	Significance
Reference (S_{ref})	0.00	27.00	Highly sensitive to very low speeds; nullifies all shared spaces.
Proposed ($Score_{prop}$)	≈ 0.11	≈ 20.09	**High-contrast**; recognizes baseline shared safety and prioritizes separation.

Table 5: Comparative extrema of the cycling safety models.

10.2 Results

Equations (2) and (8) were used to determine the Global City Index and to establish a comparison between the reference model and the proposed model.

Global City Index (Reference): 5.96
Global City Index (Proposed): 3.95

10.3 Experimental Results on Compiègne and Normalization

The implementation of the models on the Compiègne network (6,013 segments) yields two distinct global indices. However, since S_{ref} and $Score_{prop}$ operate on different mathematical ranges, a normalization step is required for a rigorous comparison.

10.3.1 Normalization Method

The reference model has a theoretical maximum $S_{ref,max} = 27.00$, while the proposed model reaches its "safety ceiling" at $Score_{prop,max} = e^3 \approx 20.09$. To compare the Proposed Global City Index (GCI_{prop}) with the Reference Global City Index (GCI_{ref}), we apply a rule of three:

$$GCI_{prop,norm} = \frac{GCI_{prop} \times S_{ref,max}}{Score_{prop,max}} \tag{13}$$

10.3.2 Comparative Results

Using the values calculated for the Compiègne network:

- **Reference Index:** $GCI_{ref} = 5.96$
- **Proposed Index (Raw):** $GCI_{prop} = 3.95$
- **Proposed Index (Normalized):** $GCI_{prop,norm} = \frac{3.95 \times 27.00}{20.09} \approx 5.31$

Analysis: Even after normalization, $GCI_{prop,norm}$ (5.31) remains lower than GCI_{ref} (5.96). This 11% decrease demonstrates that the proposed model is more discriminating. It reveals that the bikeability of Compiègne is overvalued by linear approaches because they fail to penalize the "stress points" created by the lack of physical protection on 50 km/h transit axes.

11 Scaling Up: Computational Performance and National Projection

Assessing the scalability of the model is critical for transitioning from local urban planning to national infrastructure policy. This section evaluates the computational efficiency of the processing pipeline and projects the resources required to model the entire French road network.

11.1 Local Benchmark: Compiègne

A performance audit was conducted on a standard workstation using the current Python implementation. The benchmark covers the complete pipeline, including data loading from Parquet files, velocity imputation, and the calculation of the hybrid safety scores for the 6,013 segments of the Compiègne network.

Metric	**Result**
Total Execution Time (I/O + Calculation)	0.5165 seconds
Processing Throughput	10,920 segments/second

Table 6: Computational performance metrics for the local city network.

11.2 National Projection: France

The French road network is estimated to contain approximately 25 million segments within the OpenStreetMap database. Based on the linear performance of the current algorithmic implementation ($O(N)$), we can project the time required for a nationwide assessment.

- **Total Estimated Time:** $\approx$ 38 minutes (0.64 hours).
- **Methodology:** The projection assumes a consistent processing throughput and linear scaling of I/O operations.

11.3 Computational Constraints and Optimization

While the execution time remains within reasonable limits for a national scale, the primary bottleneck is memory management (RAM). Loading the entire French network into a single GeoPandas GeoDataFrame would require significant RAM resources (estimated at 32-64 GB). To move toward a national production-ready tool, the following optimizations are recommended:

1. **Batch Processing:** Partitioning the network by administrative regions to reduce memory footprint.
2. **Parallelization:** Utilizing libraries such as `Dask` or `Spark` to distribute calculation across multiple CPU cores.

12 Discussion

The results of this study demonstrate that the proposed hybrid model offers a more nuanced and realistic assessment of urban bikeability compared to traditional linear approaches. By integrating the "psychological jump" associated with protected infrastructure and the conditional nature of traffic speed risks, the model provides a high-contrast evaluation of the cycling network in Compiègne.

12.1 The "Psychological Jump" and Non-linearity

Our first hypothesis regarding the non-linearity of safety proved essential in distinguishing between basic painted lanes and physically separated tracks. While the reference model ($GCI_{ref} = 5.96$) tends to smooth out differences, our normalized proposed model ($GCI_{prop,norm} = 5.31$) reveals an 11% decrease in the overall assessment. This suggests that a linear model overestimates the safety of a city by not sufficiently penalizing infrastructure that fails to meet the threshold required to attract the "interested but concerned" demographic.

12.2 Neutralizing Speed through Separation

The implementation of the kinetic weighting factor $\delta(V, i)$ validates our second hypothesis: speed is only a dominant stress factor when the cyclist is exposed. On segments where physical separation is present ($i \geq 2$), the model correctly neutralizes the speed penalty, reflecting a real-world perception where a cyclist feels safe even adjacent to 50 km/h traffic if a barrier exists. Conversely, the severe penalization of shared spaces in 50 km/h zones highlights "critical gaps" that are often invisible in standard models but represent major barriers to modal shift.

12.3 Implications for Urban Planning

From a decision-making perspective, the "high-contrast" mapping produced by the hybrid model serves as a strategic tool for urban planners. Rather than distributing resources thinly across numerous painted lanes, the model suggests that investing in a continuous "cycling backbone" of protected infrastructure is more effective for increasing the Global City Index and ensuring user safety.

12.4 Limitations and Future Work

While the current model focuses on safety, the most dominant criterion for bikeability, it does not yet fully integrate other factors such as pavement roughness (IRI), network directness, or intersection complexity. Furthermore, the reliance on OpenStreetMap data necessitated significant velocity imputation (82% of segments), which, while based on expert standards, introduces a layer of estimation. Future iterations could incorporate real-time traffic flow data and accident history to further validate the mathematical weightings of the exponential and kinetic factors.

13 Conclusion

This work set out to build a transparent and reproducible framework for assessing urban bikeability using open data. By moving beyond traditional linear scoring methods, the study introduced a model capable of capturing the sharp contrasts that cyclists actually experience on the ground. The approach highlights how infrastructure quality and exposure to motorized traffic interact in ways that simple additive models fail to represent.

Applied to the city of Compiègne, the model reveals a network whose strengths and weaknesses become far more visible once non-linear safety perception and conditional speed effects are taken into account. Rather than offering a uniform picture of the city, the results expose the segments where safety breaks down and where continuity of protected infrastructure would yield the greatest improvements. This makes the model not only a measurement tool but also a strategic instrument for prioritizing interventions.

While the present work focuses on safety-related factors, the methodology opens the

door to broader extensions, such as integrating comfort, connectivity, or user perception data. Future developments could also compare cities, validate the model against real-world usage patterns, or refine the treatment of speed and separation. Nonetheless, the findings already demonstrate that a more nuanced mathematical formulation provides a clearer and more actionable understanding of urban bikeability.